

SafeOT-Dual

Penalty view

Costs enter objective

$$C_k \longrightarrow \lambda_k \longrightarrow \Sigma$$

exact settings

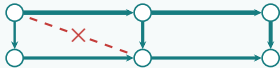
$c = 0, \beta = \text{const}$

Fixed penalty

$c = 0, \text{adaptive } \beta$

Lagrangian

Solve one graph problem



budgets b_k · flow balance · KL

Price

$$\lambda_k = \text{clip-filter}(\lambda_k^*) \oplus \beta_k$$

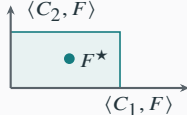
feedback: $\bar{J}_k - B_k$

Update the actor

$$\tilde{A} = \frac{Ar - \sum_k \lambda_k A c_k}{1 + \sum_k \lambda_k} \rightarrow \text{PPO/TRPO}$$

Constraint view

Budgets bound graph costs



limits

$\varepsilon \rightarrow 0$: LP dual price

$\varepsilon \rightarrow \infty$: clip saturates

active-set price

(under assumptions)